Applied Algebra

Spring 2026

Instructor: Tuan Tran

Exercise Sheet 6

Deadline: by the end of next Wednesday

Instructions. You are encouraged to work in pairs. When submitting your solutions, list the team members and circle the name of the person responsible for writing up the solutions that week.

Please submit written solutions to **TWO** of the following problems. **Justify your answers.**

Problem 1. For each of the following relations R on the set X, decide whether R is reflexive, symmetric, antisymmetric, or transitive.

(a) $X = \mathbb{Z}$, aRb if and only if $a \leq b + 1$;

(b) $X = \mathbb{N}$, aRb if and only if a and b are coprime;

(c) $X = \mathbb{Z}$, aRb if and only if $a + b$ is divisible by 3;

(d) $X = \mathbb{N} \times \mathbb{N}$, $(a, b)R(c, d)$ if and only if either $a < c$ or $a = c$ and $b \leq d$.

Problem 2. Let $X = \{1, 2, 3, 4\}$ and let

$$R = \{(1, 1), (1, 2), (2, 1), (2, 2), (3, 3), (3, 4), (4, 3), (4, 4)\}.$$

Write down the adjacency matrix of R. Show that R is an equivalence relation, and determine its equivalence classes.

Problem 3. Draw state diagrams for the following finite state machines, and determine the words accepted by the corresponding automata.

(a) The machine M with $S = \{0, 1, 2\}$, $A = \{a, b\}$, and μ is given by

	a	b
0	0	1
1	1	2
2	2	0

with accepting set $F = \{1\}$.

(b) The machine M with $S = \{0, 1, 2\}$, $A = \{a, b, c\}$, and μ is given by

	a	b	c
0	1	0	2
1	0	0	1
2	2	0	2

with accepting set $F = \{2\}$.

Problem 4. Design finite state machines to meet the following specifications.

(i) An automaton with alphabet $\{a, b\}$ that accepts only those words consisting entirely of the letter a.

(ii) An automaton with alphabet $\{0, 1, \ldots, 9\}$ that accepts only the four-digit word 1357.